

Unit-8 $\Rightarrow$ photogrammetry :-

Introduction :-

- Derived From Greek word -
- photo - Means "Light".
- gram - Means "Draw something" or "write something".
- metron - Means "Measurement". Thus,

"Photogrammetry is defined as the science of obtaining reliable measurements, by the use of photographs, in-order to determine characteristics, shape and position of photographed objects. (i.e. Art of producing map by compiling photograph is called photogrammetry). There are two techniques of taking photographs of surveying area. a) By Air or b) Elevated ground stn.

1) Aerial photogrammetry -

The photographs are taken by a camera mounting in an aircraft flying over an area with constant speed and constant time interval by overlapping forward as 60% and side by 20%.

2) Elevated ground stn. (Terrestrial photogrammetry) :- photographs are taken from Elevated ground station by using photo theodolite. (Theodolite fitted with the camera).

The principle of Terrestrial photogrammetry is exactly similar to that of plane table survey by radiation method. Due to Less Accuracy not being used in Engineering survey.

1) Aerial photogrammetry :-

- Extensively used in Engineering, Forestry, geology, Agriculture, urban planning etc.
- photographs are taken from camera station in air.
- Axis of camera vertical or nearly vertical.

According to Axis of camera - Aerial photographs are of two types -

1) vertical photograph :-

Since, vertical photographs are most commonly used in modern Aerial photogrammetry and photo-interpretation, the term photograph refers to a vertical Aerial photograph.

- Taken from Air with camera Axis vertical or nearly vertical.
- Axis of camera vertical $< 2^\circ$.
- Used for topographical and Engineering survey on various scale.
- A Truly vertical photograph closely Resembles a map.

2) Oblique photograph :-

The photograph taken from the air by tilting camera axis intentionally from vertical is called oblique photograph.

- Axis of camera inclined from vertical line $> 2^\circ$.
- Covers Large area.
- Further Divided into 1) Large oblique - (High oblique)
2) Low oblique $\leq 20^\circ$ from v-axis
High oblique $\geq 20^\circ$ from v-axis

⇒ scale of vertical photograph:-

Scale is defined as the Relation ship between a Linear distance on a vertical photograph and the corresponding actual distance on the ground.

$$\therefore \text{Scale} = \frac{\text{Linear Distance on photo}}{\text{Corresponding Linear Distance on ground}}$$

Corresponding Linear Distance on ground

- photographic scale indicates proportional distance. ✓

Example- Scale - 1:15,000 means that length of 1 unit of measurement on the photo represents 15,000 units of measurement on ground.

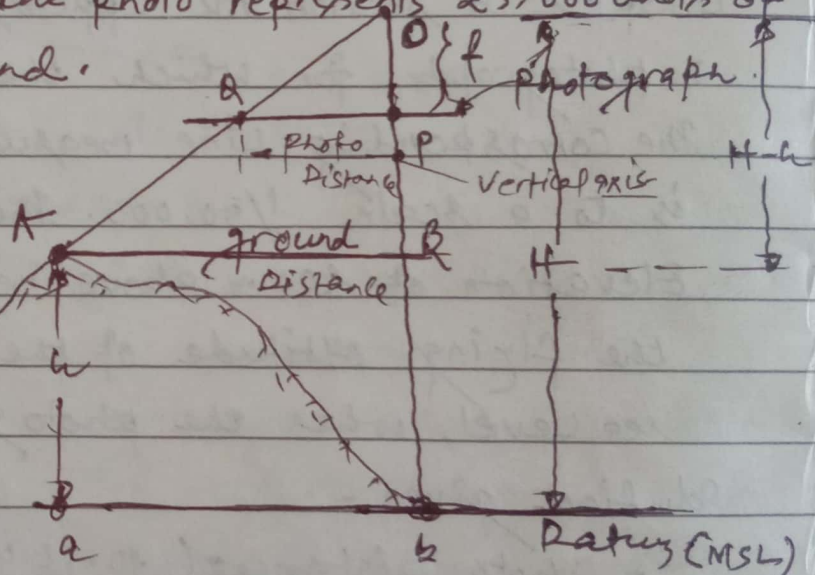
In fig -

Let, O is the Perspective centre.

origin of the lens.

Exposure station of the flight.

H = Flying height of the camera.



A = hill top at height 'h' from MSL.

OB = vertical axis.

Q = Edge of Photograph.

f = focal length of the camera.

PROVE From similar triangle $\triangle OAB$ and $\triangle OPQ$.

$$\frac{PO}{OB} = \frac{PQ}{AB} \quad \text{OR,} \quad \frac{f}{H-h} = \frac{\text{Photo Distance}}{\text{Ground Distance}}$$

$$\therefore \text{Scale} = \left(\frac{f}{H-h} \right)$$

- For plane topography, $h \approx 0$

$$\text{Scale} = \left(\frac{f}{H} \right)$$

In case of Reliable map of ^H the area is available photo graphic scale can be found by comparing the photo distance and the map distance betⁿ two well defined points at the same Elevation -

$$\text{is } \frac{\text{Photo scale}}{\text{Map scale}} = \frac{\text{Photo Distance}}{\text{Map Distance}} \quad \checkmark$$

Example - A section AB appears to be 10.16 cm on a photograph for which the focal length is 16 cm, the corresponding line measures 2.54 cm on a map which is to a scale $1/50,000$. The terrain has an average Elevation of 200 m above mean sea level. Calculate the flying altitude of the air-craft, above mean sea level, when the photograph was taken.

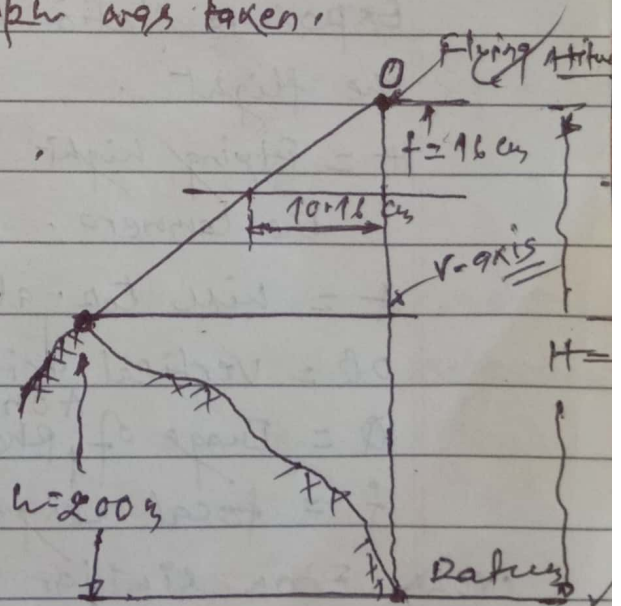
Solution - given -

- Photo Distance = (10.16 cm)
- Focal length = 16 cm
- Map Distance = 2.54 cm
- Scale of map = $1/50,000$

Then, - we know, -

Relation betⁿ the photo scale and map scale is

$$\text{given by} - \frac{\text{Photo scale}}{\text{Map scale}} = \frac{\text{Photo Distance}}{\text{Map Distance}}$$



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$$\therefore \text{photo scale} = \text{map-scale} \times \frac{\text{photo distance}}{\text{map distance}}$$

$$= \frac{1}{50,000} \times \frac{10.12}{2.54}$$

$$= \frac{1}{12,500} \quad \therefore \underline{1:12500}$$

Again, we have, -

$$\text{photo scale} = \frac{f}{H - h}$$

$$\text{or } \frac{1}{12500} = \frac{16 \text{ cm}}{H - 200}$$

$$\text{or } H - 200 = 0.16 \times 12,500$$

$$H - 200 = 2000$$

$$\text{or Flying height (H)} = 2000 + 200 = 2200 \text{ m} \quad \checkmark$$

Example: (2) A line 2250 m long lying at an elevation of 500 m, measures 10.50 cm on vertical photograph. The focal length of the camera used is 20 cm. Determine the scale of the photograph for an area having an elevation of 1200 m.

Solution - given -

$$h = 500 \text{ m}$$

$$\text{Ground Distance} = 2250$$

$$\text{Photo Distance} = 10.50 \text{ cm}$$

$$f = 20 \text{ cm}$$

$$\text{photo scale having elevation } 1200 \text{ m} = ?$$

At 500 m -

$$\text{photo scale} = \frac{\text{photo distance}}{\text{ground distance}}$$

$$\left(\frac{10.50}{2250} \right) = \frac{f = 20 \text{ cm}}{H - 500}$$

$$\text{or } H - 500 = \frac{2250 \times 20}{10.50}$$

$$\therefore H = 2023.0710$$

2023.07.10 10:37

the fore, scale at 1200 is -

$$\text{Scale} = \frac{2003 (0.20^m)}{(4976.20 - 1200)}$$

$$\therefore \text{Scale Rf} = \frac{1}{18881}$$

⇒ Relief Displacement :-

(Height of Displacement) -

Since the photograph is the ~~pers~~ perspective view, the ground height is shown in perspective on the photograph.

Every point on the photograph is therefore displaced from their true orthographic position.

This displacement is

called Relief Displacement.

- To calculate Amount of Relief Displacement - Consider above fig -

Let -

r = Radial Distance q' from (q' - Principle point)

r_0 = Radial Distance q_0 from (q_0 - Principle point)

$R = VA_0$

Then -

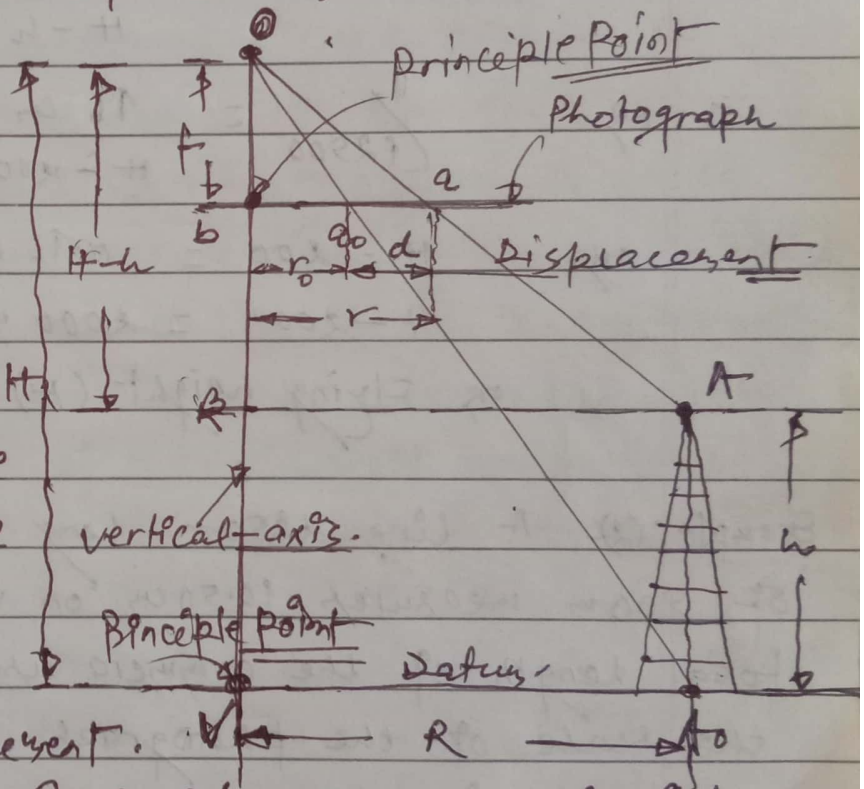


Fig:- Relief Displacement

Then

From similar ΔOab and ΔOAB .

$$\frac{Ob}{Oa} = \frac{ab}{AB}$$

$$\text{i.e. } \frac{f}{H-h} = \frac{r}{R}$$

$$\text{or, } (r) = \left(\frac{Rf}{H-h} \right) \quad \text{--- (7)}$$

Also From similar ΔOab and $\Delta OVAo$.

$$\frac{Ob}{OV} = \frac{aob}{VAo}$$

$$\text{or, } \frac{f}{H} = \frac{r_o}{R}$$

$$\text{or, } (r_o) = \left(\frac{Rf}{H} \right) \quad \text{--- (8)}$$

Therefore from Eqn (7) and Eqn (8).

Relief Displacement (d) is given by -

$$\text{Relief Displacement } (d) = r - r_o$$

$$= \frac{Rf}{(H-h)} - \frac{Rf}{H}$$

$$= \frac{RfH - RfH + Rfh}{H(H-h)}$$

$$(d) = \frac{Rfh}{H(H-h)} \quad \text{--- (9)}$$

Also, From Equation (i) -

$$R = \frac{r(H-h)}{f}$$

From Equation (ii),

$$R = \frac{r_0 H}{f}$$

Now substitute the value of R in Eqn (iii) we get

$$d = \frac{r(H-h)/f \times h}{f(H-h)/H}$$

$$\therefore \left[d = \frac{rh}{H} \right] \checkmark$$

Also, $d = \frac{r_0 H}{f} \times \frac{f h}{H(H-h)}$

$$\therefore \left[d = \frac{(r_0 h)}{H-h} \right] \checkmark$$

From the above Equation conclusion is -

- 1) When Relief Displacement increases as the Distance from principle point increases -
- 2) Relief Displacement Decrease with increase in Flying height.
- 3) For point above Datum Relief Displacement is positive being Radially outward -
- 4) For point below Datum (h having (-ve) value) Relief Displacement negative radially inward -
- 5) The Relief Displacement of point vertically below the Exposure station is zero.

Unit:-9:- Astronomy and GPS:-

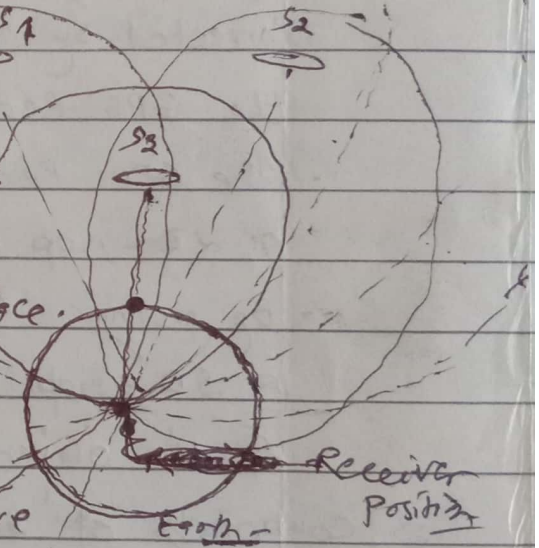
Introduction to Global positioning system (GPS):-

The global positioning system (GPS) is defined as the radio navigation system involving satellite and computers that can determine the latitude and longitude of a receiver on the earth by computing the time difference for signals reaching from different satellites to the Receiver.

GPS is used to support a broad range of military, commercial and consumer applications.

There are almost 30 GPS satellites ^{S1} out of which 27 satellites are active and rest are spare, ~~sat~~ situated in 6 orbits at a height of 10600 miles above the earth surface.

The position of satellites are such that for any point on the earth, Every four satellite will be above horizon.



The GPS-satellites contains a Computer, an atomic clock and a Radio. Each satellite continuously broadcasts its changing position with time to the Receiver on the earth. The Receiver contains a Computer, which triangulate its own position by getting bearings from three of the four satellites. So the exact location of the Receiver at a specific time instant can be determined in terms of Latitude and Longitude.

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- If the receiver contains the Display screen, the position is displayed over the map with respect to time. The moving receiver can also determine the speed, direction of translation and the estimated time to reach a specified destination.

⇒ Functioning of GPS :- (Working of GPS) :-

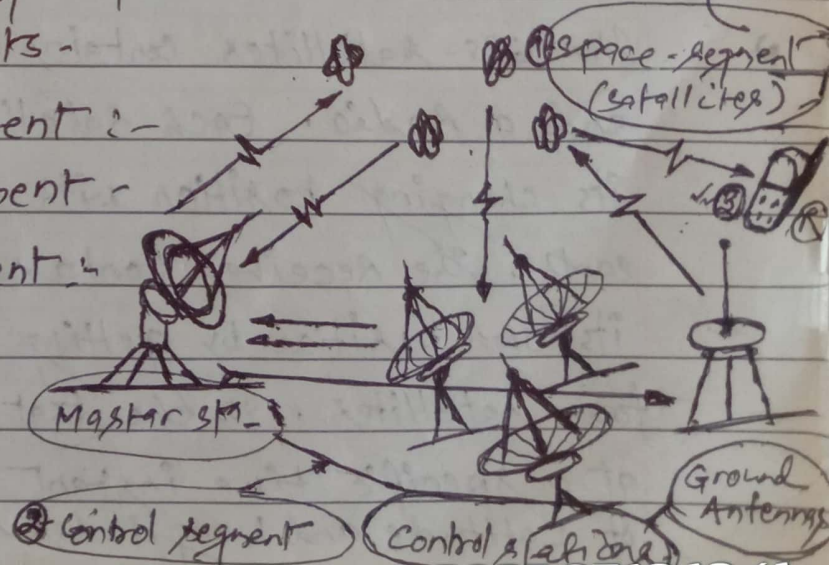
The GPS satellites are orbiting the earth continuously. The Radio signals from satellites are controlled and corrected by control station. signals are received by the GPS Receiver on the earth.

- The GPS Receiver needs only 3-satellites to plot a 2D-map.
- By using 4-satellites it can correctly draw a 3D-map to locate the geographical position of the object over the earth surface.

⇒ Components of GPS :-

The entire functioning of GPS is carried out by three main components -

- 1) Space segment :-
- 2) Control segment -
- 3) User segment :-



1) Space-segment :-

- The space segment consists of 30-GPS-satellite inclined at 55° and orbiting around in Every 12-hrs from height of 10660 miles above the Earth surface.
(Due to earth's rotation on its own axis, a satellite will take 24-hrs for a complete Rotation around earth).
- The position of GPS-satellites are such that every four satellite covers a specified point (Receiver) on the earth surface. (The satellite signal can be
- Received any where within the satellites effective Range).
- The Low Radio signal emitted continuously from satellites has a definite frequency for allowing the receiver to identify the signal.
(The signal moves at a speed equal to that of light)
- The time elapsed for reaching the signal from the satellite to receiver can ultimately determine the Distance of the Receiver from the corresponding GPS-satellite.

2) Control segment :-

- The control segment consists of five unmanned monitor stations and one master station.
- The monitor station continuously receive Radio signals emitted by the GPS-satellites and transmitted to the master station for its necessary correction on time

and orbital location.

- The corrected information is then sent back to the GPS-satellite through the ground stations antennas.

8) User Segment.

- The user segment consists of the user and their GPS-Receiver and its no. is numerous.
- The signal transmitted by the satellites takes approximately 67 microseconds to reach a Receiver.
- Four different signals are generated in the Receiver having the same structure as those received from the four satellites.
- By ~~the~~ synchronising the signal generated, the four signal time shifts " Δt " are measured as a timing mark. The time shift for all four signals are used to determine the signal transit time.
- The signal transit time is used to determine the distance of respective satellites. The Receiver then calculates the Latitude, Longitude, ~~and~~ height and time of the user from the known range of four satellites://

→ Uses of GPS Application -

1) Location :-

GPS is used for finding location of any object lying on ground, on sea surface or in air at any time in any weather condition.

2) Navigation :-

GPS helps us to determine exactly where we are. But sometimes it is more necessary to know how to get somewhere else. GPS was originally designed to provide navigation information for aeroplanes and ships.

3) Tracking :-

GPS technology is also useful for monitoring vehicles and persons, Tracking is useful in the following matters - Mass Transit -

- Ship Tracking -

- Vehicle Tracking -

GPS Tracking is widely used by POLICE, Ambulance and Fire Department.

4) Mapping and Survey :-

- By using GPS map regarding mountains, Rivers, Towns etc are prepared -

These maps are useful for :-

1) Conservation of Natural Resources.

2) Forest monitoring.

3) Weather forecasting.

4) Geodetic surveying - ✓

Unit-10 Remote sensing and GIS:-

Introduction to Remote sensing:-

Remote:- Something which is far away -

sensing:- Getting information or getting Data -

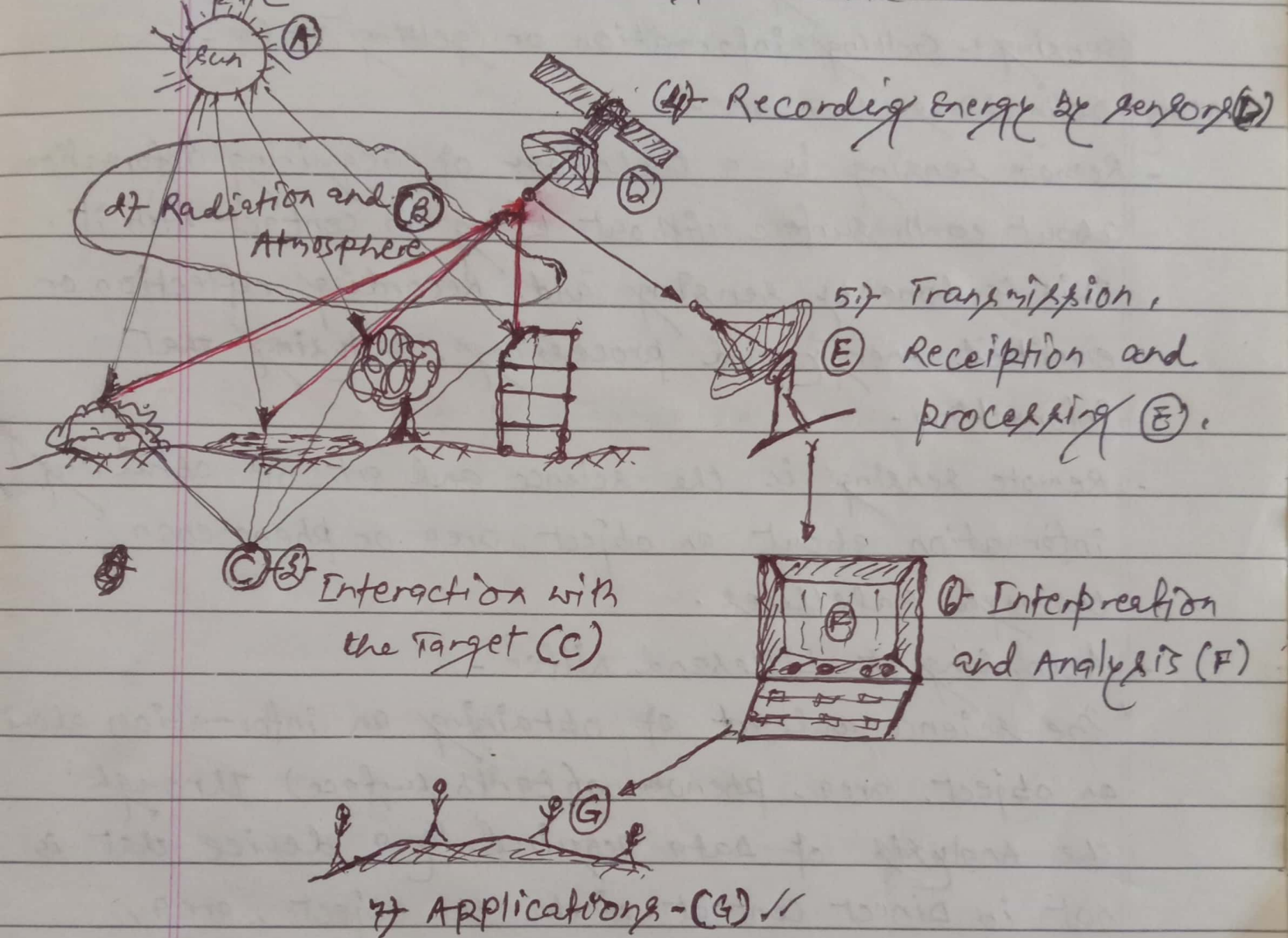
Many definitions some are -

- Remote sensing is a technology of acquiring information about earth surface without being in contact with it. This is done by sensing and Recording reflection or emitted energy and processing, analysing that information.
- Remote sensing is the science and art of obtaining information about an object, area or phenomenon through satellites.
- According to Lillesand and Kilfer -
"The science and art of obtaining an information about an object, area, phenomena (Earth surface) through the Analysis of Data acquired by a device that is not in direct contact with the the object, area, phenomena under the investigation",
- The science and art of acquiring information about an object without entering in contact with it, by sensing or Recording the Reflected or emitted energy and processing, analysing and Applying that Information is Remote sensing.

→ Remote Sensing :-

How it works :- (ie process of Ideal Remote sensing) :-

1) Energy source (A) - Electromagnetic Energy source -



1) Uniform Energy source (A) :-

The first Requirement for Remote sensing is to have an Energy source which illuminates or provides electromagnetic energy to the target interest.

(Source - may be sun or other emitted Electromagnetic energy)
Electromagnetic waves, Microwaves etc).

2) Radiation and Atmosphere:- (A) :-

As the Energy Travels from its source to the target, it will come in contact with the atmosphere it passes through; which will not modify the Energy transmitted from the source or emitted (Reflected) from the object in any manner.

3) Interaction with the Target: (C) :-

A series of unique Energy transmitted by the source interacts with the target, it reflects or emits signals that are selective with respect to wavelength and also unique to ~~each~~ each object or earth surface feature type.

4) Recording of Energy by the sensors:- (D) :-

After Energy has been scattered by, emitted from target, we require sensor which is highly sensitive to all wavelengths to collect and record the Electromagnetic Radiation (Energy).

5) Transmission, Reception and processing (E) :-

The energy required by the sensors has to be transmitted, often in electronic form, to receive and processing section, where the data are produced into an image (Hard copy and Digital).

6) Interpretation and Analysis : (F)

The produced image is interpreted in two ways, visually or Digitally. To extract information about the target which was illuminated.

7) Application :- (G) :-

The final element of Remote sensing process is achieved when we apply the information - we have able to extract from the image about target in order to better understand and Reveal some new information. ✓

⇒ Components of Remote Sensing :-

There are three main components of Remote sensing

1) platforms :-

2) Sensors :-

3) orbits :-

1) Platforms :-

A platform is the vehicle or carrier for Remote sensors. There are three types of platforms in Remote sensing -

1) Ground base platforms - sensor ≥ 50 km from Earth

2) Airborne platforms - sensor above 50 km

3) Space born platforms - ~~up to~~ ≥ 50 km - ✓✓

2) Remote sensing sensors :-

Device that Receives Electromagnetic Radiation and Converts it to a signal that can be Recorded and Displayed as either numerical data or an image -

There are two types of sensors -

1) Active sensors -

2) Passive sensors -

1) Active sensors :-

It transmit their own signal and measure the energy that is Reflected and transmitted back or scatter back from the target. For Example - Flash of a camera -

2) Passive sensors :-

It detect natural energy (Radiation) that is emitted or Reflected by the object or scene being observed. Reflected sunlight is the most common source of Radiation measured by passive sensors.

2) Orbit :-

The curved path of the sensor carrier (satellite) Revolves the Earth in the Atmosphere - Orbits are -

1) Low Earth orbit (LEO) :-

2) Medium Earth orbit (MEO) .

3) Geosynchronous Earth orbit (GSO) and

4) Geostationary Earth orbit (GEO)

⇒ Application of Remote sensing -

In the Remote sensing process the satellite sensors are continuously monitoring the Earth's natural and other Resources and of Determining the impacts of man's activities on air, water and Land. Therefore, some specific Applications of Remote sensing is given below -

- 1) Land-use mapping -
- 2) Forest and Agriculture Applications -
- 3) Environmental Applications -
- 4) Hydrology and Coastal mapping -
- 5) Urban planning -
- 6) Natural Resources based Application -
- 7) Engineering geological studies -
- 8) Drainage Analysis -
- 9) Geo morphological mapping -
- 10) Impact assessment on vegetation, water bodies.

⇒ Advantages and Disadvantages -

Ref. Engineering Surveying. I and II

En: Bharat Bdr. Dhalal -

- Introduction to GIS:-

A Geographic Information system refers to a system of capturing, storing and manipulating spatial information in Digital form ~~which~~ with the help of the computer hardware and software to analyse and present the features of the Earth. Actually no specific definition of GIS. Some of them are defined according to the purpose of its use -

" A GIS is defined as a computer based system that provides the following four set of capabilities to handle geo-referenced data -

1) Input -

2) Data management (Storage and retrieval) -

3) Manipulation and analysis and

4) out put -

The GIS is basically a system by which spatial data in digital form is analysed, manipulated and displayed with the aid of investigation of computer software and hardware -

⇒ Components of GIS:-

A successful GIS operation needs following components

1) GIS - Hard ware :-

The hardware in GIS means a computer having large memory space of acquiring database with highly

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sophisticated displaying facility. For the most cases for combination of different data references, more than one computer is used. The computers are connected to a common server during operation -

2) GIS-Software :-

The GIS-Software is used to collect, analyse and display the spatial information. Software actually plays the key role for manipulation in the case of data collected in different scales and projections.

3) Data :-

Data is the backbone of GIS. The spatial data and tabular data are combined to form a data structure of GIS. Actually, there are two types of data in a GIS.

1) Spatial Data :-

Collected ~~from~~ from satellite monitoring, photography or other similar type of linkage in the form of digital image system.

2) Attribute Data :-

It describes geographic regions or defines characteristics of spatial features within the geographic region. ✓

4) people :- User :-

The success ~~of~~ of the GIS is judged by the end users. Some of the users are interested to use its technical aspects, and a respectable number of people use it for solving their Real-life problems. The co-relation betⁿ. professional GIS end users are very important for successful implementation of GIS.

5) Methodology :-

The well-defined methodology with an integrated approach of GIS is Required. Hiring of correct hardware and software for different types of GIS ~~are also very important for GIS field~~ and Extracting data appropriately for the purpose is an important part of this methodology.

⇒ Applications :-

A GIS enable us to better understand and evaluate data by creating graphic display using information stored in our data base. The Major Application of GIS can be grouped in to 5 Categories -

- | | |
|-------------------------------|----------------------|
| 1) Facilities Management - | 4) Street Networks - |
| 2) Planning and Engineering - | 5) Environment and |
| 3) Land Information system - | Natural Resource |